	Registration (live → fixed)	Host / nucleus segmentation	TUNEL call	Parasite localisation	Measurement & statistics	Time T1 / T2	Task 1 TUNEL calls (6 nuclei, fixed frame)	Task 2 published Fig. 4C (15 parasites, last live frame)
Published	by hand	hand-drawn ROIs	by inspection	15 ROIs + 3 backgrounds each, by hand	ROI Manager → Excel; Mann–Whitney	not recorded	2/6 nuclei TUNEL+	8 TUNEL+ vs 7 TUNEL−, p = 0.0012
Codex CLI	SIFT + RANSAC, ECC refined (1.6 px)	T1 DAPI threshold; T2 Cellpose cyto3	largest-gap cutoff	red-channel peaks + water-shed: 10 kept	annulus background; no test	18 min (both)	✓ 6/6 calls, r = 0.98	✓ 10/15 found, hosts right; p = 0.017
Exploratory Agentic-J	TurboReg rigid on brightfield	T1 StarDist; T2 Cellpose cyto3 (7 hosts)	largest-gap cutoff	C2 + C3 spot evidence: 22 kept	annulus background; no test	22 / 69 min	✓ 5/6 calls, r = 0.95, 1 split	✗ 0/15 found (+22), no host
Conventional Agentic-J	TurboReg / StackReg rigid	T1 DoG + water-shed; T2 Cellpose cyto3 (6 hosts)	2-component GMM	micro-sam vit_b_lm: 105 → 5 kept	annulus background; no test	55 / 85 min	✓ 5/6 calls, r = 0.99	✗ 3/15 found (+2), no TUNEL+
Conventional Agentic-J + human	lab ROIs given, origin translated	lab ROIs (given)	largest-gap cutoff	lab ROIs paired 1:1 with backgrounds	native 16-bit; Mann–Whitney	32 min / n.r.	✓ 5/6 calls, r = 0.97	✓ 15/15; 14/15 within 0.1; p = 0.0012

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